Hamed Khosravi

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Research Profile

Machine Learning Ph.D. student working on **online learning under non-stationarity**, reinforcement learning, statistical learning, and sequential decision-making. My research focuses on how learning systems adapt as data distributions, latent structure, and model behavior evolve over time, with recent work on non-stationary contextual bandits, sequential evaluation, and reliability of adaptively updated LLMs. I am broadly interested in the theoretical and empirical foundations of **adaptive learning in changing environments**, including continual learning, representation adaptation, uncertainty, and optimization under sequential feedback.

Education

Georgia Institute of Technology Atlanta, GA
Ph.D. in Machine Learning, GPA: 4.0/4.0 Aug. 2025–Present
Advisor: Prof. Xiaoming Huo Anticipated Graduation: May 2028

West Virginia University Morgantown, WV
M.S. in Industrial Engineering (Data Science), GPA: 4.0/4.0 2022–2025

Relevant Coursework: Deep Reinforcement Learning, Theory of Statistics, Machine Learning, Mathematical Foundations of Machine Learning, Convexity, Advanced Nonlinear Programming

Selected Research

Conformal Selective Acting: Anytime-Valid Risk Control for RLVR-Trained LLMs **NeurIPS 2026**
Hamed Khosravi, Xiaoming Huo [Paper] [Code]

- Studied reliability during **sequential LoRA-based RLVR updates of deep transformer networks**, developing anytime-valid risk control as LLMs adapt under changing data and distribution shift.

Catching a Moving Subspace: Low-Rank Bandits Beyond Stationarity **Under review, ICLR 2027**
Hamed Khosravi, Xiaoming Huo [Paper] [Code]

- Developed **low-rank contextual bandits under non-stationarity**, combining structured estimation and change detection to adapt as latent subspaces and user preferences evolve over time.

Rank Confidence Sequences: Anytime-valid Leaderboards **Under review, ICLR 2027**
Hamed Khosravi, Xiaoming Huo [Paper] [Code]

- Developed anytime-valid sequential inference for **continual model evaluation and ranking**, allowing uncertainty to remain valid as evaluation data accumulate adaptively.

Ongoing Research

Foundation Bandit: Pretrained Sequence Models for Adaptive Decision-Making *Ongoing*

- Developing a pretrained sequence model for **continual adaptation and non-forgetting** across changing bandit tasks, with an independent statistical safety layer.

World Models for Efficient Robot Policy Selection *Ongoing*

- Studying how world-model predictions can guide which expensive real-world robot evaluations to perform while preserving statistical guarantees on final policy selection.

Industry Experience

Machine Learning Engineer Intern **Tesla**
Cell Manufacturing – Analytics & Software Team Jan.–Aug. 2025, California

- Worked in an open-ended research and engineering setting: investigated production data and engineering processes to **identify high-impact ML problems**, formulated statistical approaches, designed analyses and experiments, and implemented reusable solutions.
- Built ML workflows spanning classification, regression, time series, anomaly detection, feature engineering, model evaluation, and uncertainty; deployed pipelines with Kubeflow, Kubernetes, Docker, Python, and SQL.

Technical Skills

Programming	Python, SQL, R, Bash
Deep Learning	PyTorch, Hugging Face Transformers, PEFT/LoRA, representation learning
Learning	Reinforcement learning, contextual bandits, online learning, non-stationary learning, multi-agent RL
Theory / Statistics	Sequential inference, uncertainty quantification, change detection, statistical learning, optimization, causal/off-policy evaluation
Systems	Kubeflow, Kubernetes, Docker, Git, pytest